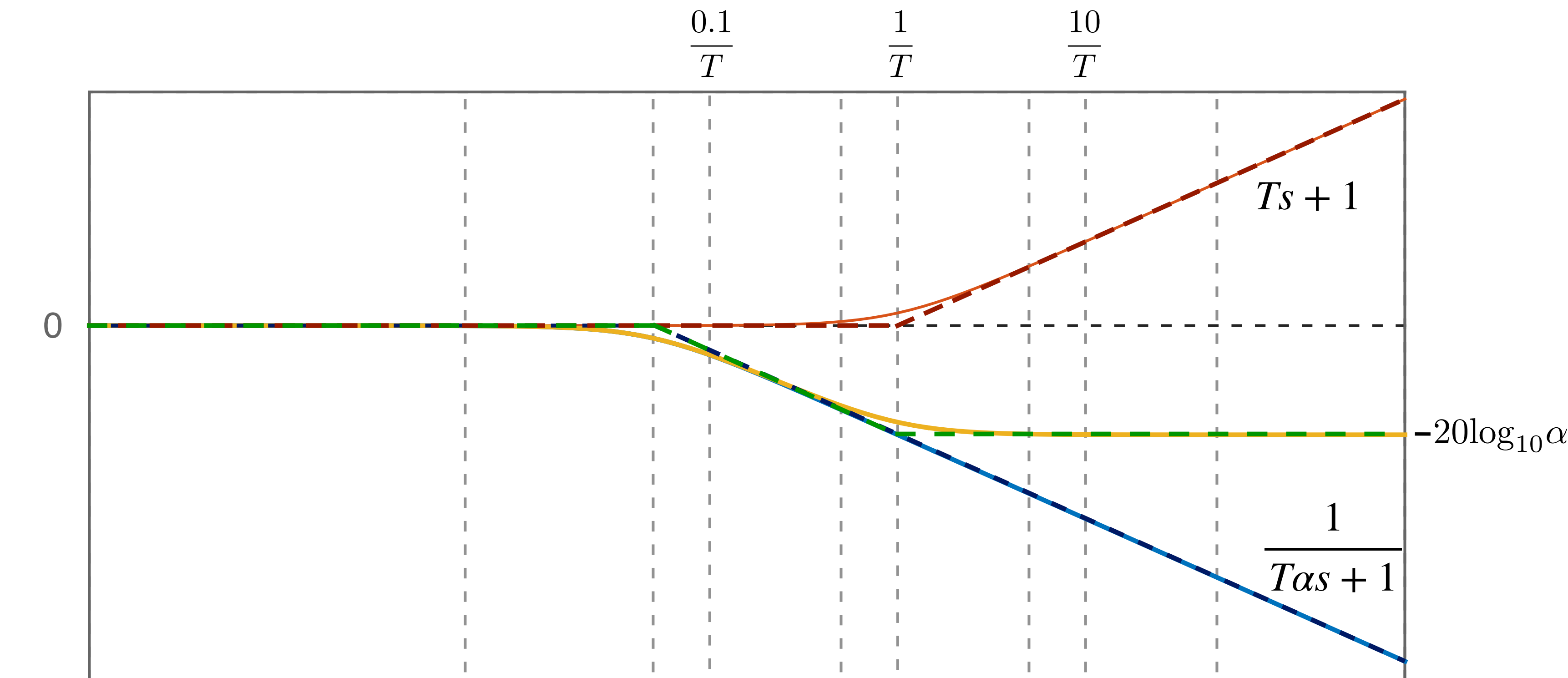


Phase-Lag Compensator

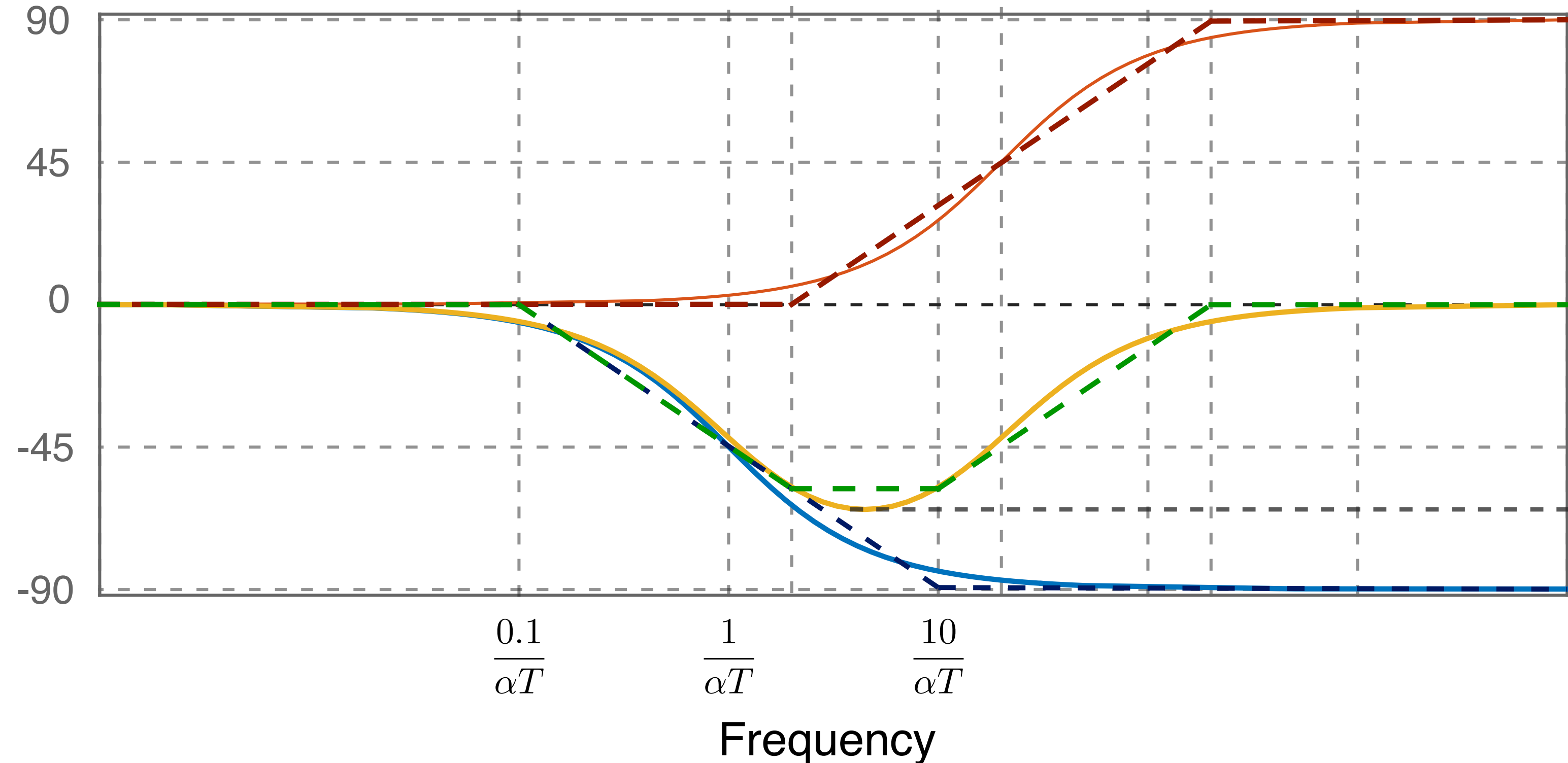
$$G_c(s) = K_c \frac{Ts + 1}{T\alpha s + 1} \quad \alpha \in (1, \infty)$$

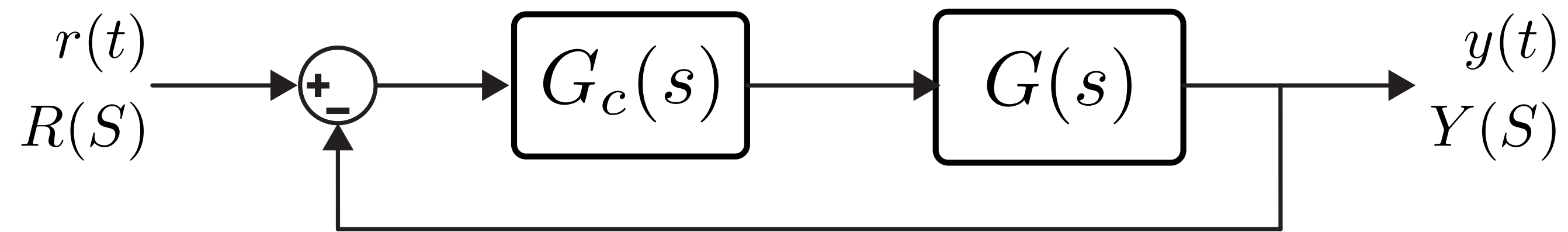
$$G_c(s) = K_c \frac{Ts + 1}{T\alpha s + 1}$$

Magnitude (dB)



Phase (deg)





$$G(s) = \frac{1}{s(s+1)}$$

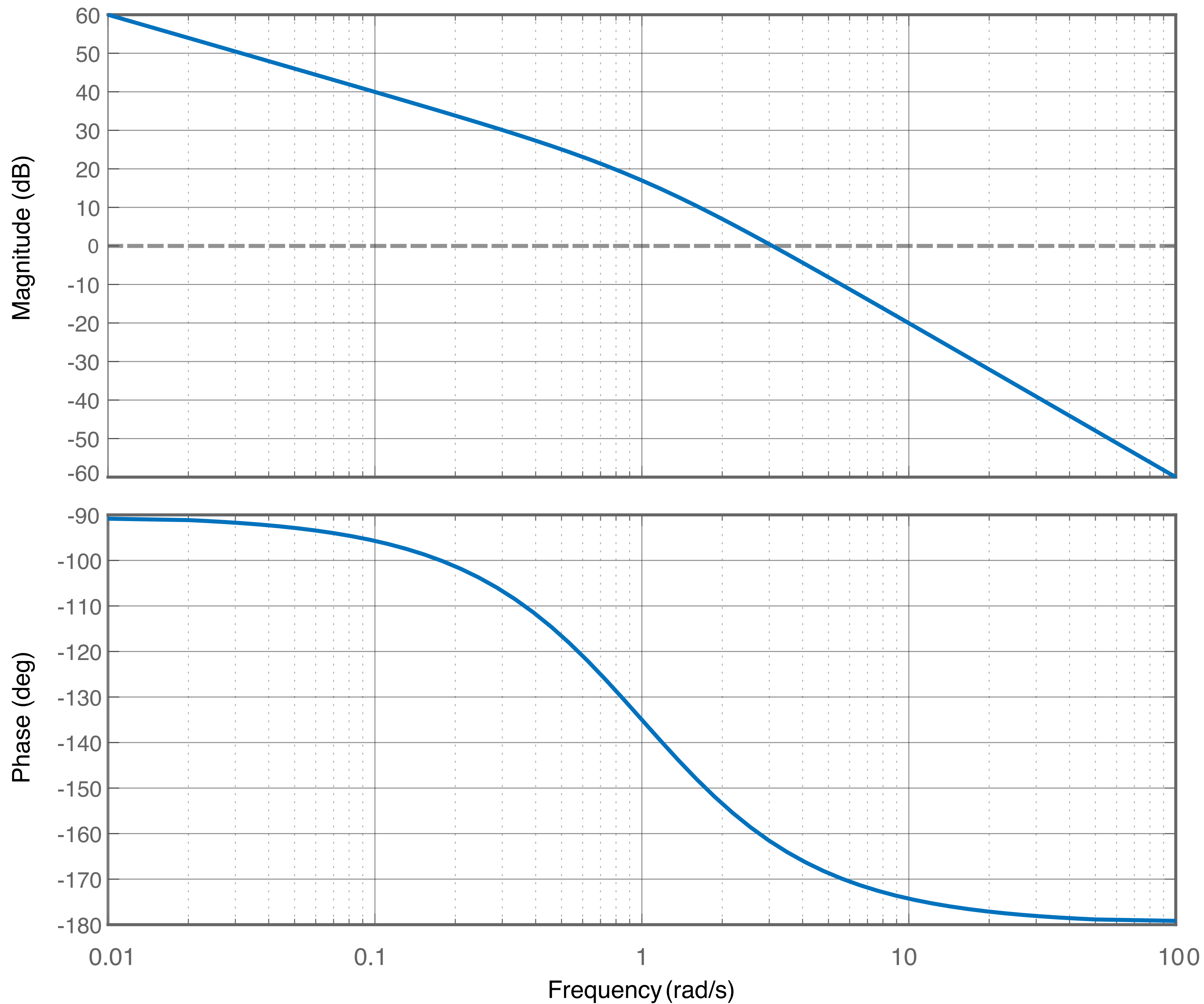
$$G_c(s) = K_c \frac{Ts+1}{T\alpha s+1} \quad \alpha \in (1, \infty)$$

Requirements

Unit-ramp steady-state error: $e_{ss} = 0.1$

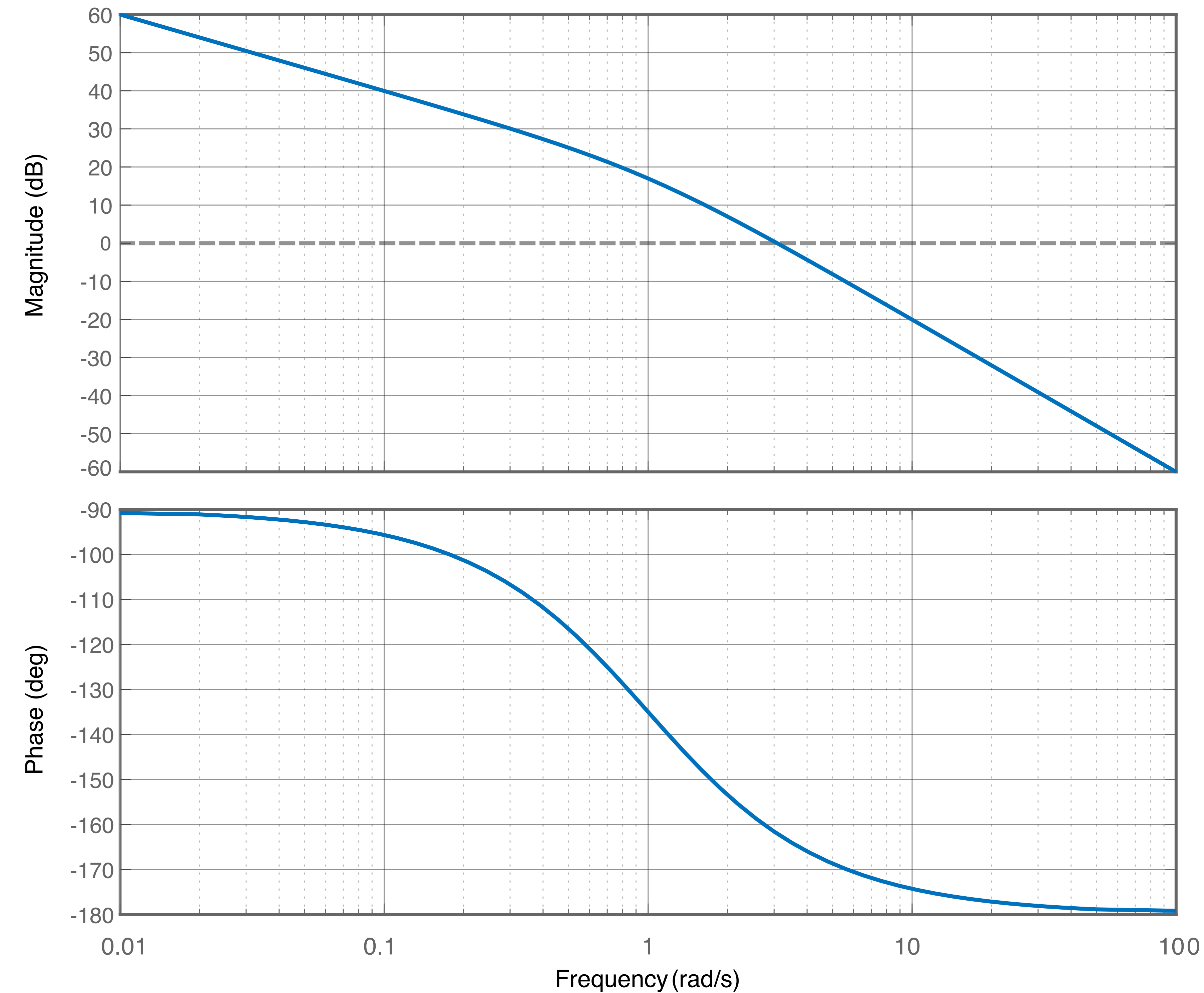
Phase margin: $\phi_M^* \approx 45^\circ$

$$\bar{G}(s) = \frac{10}{s(s+1)}$$



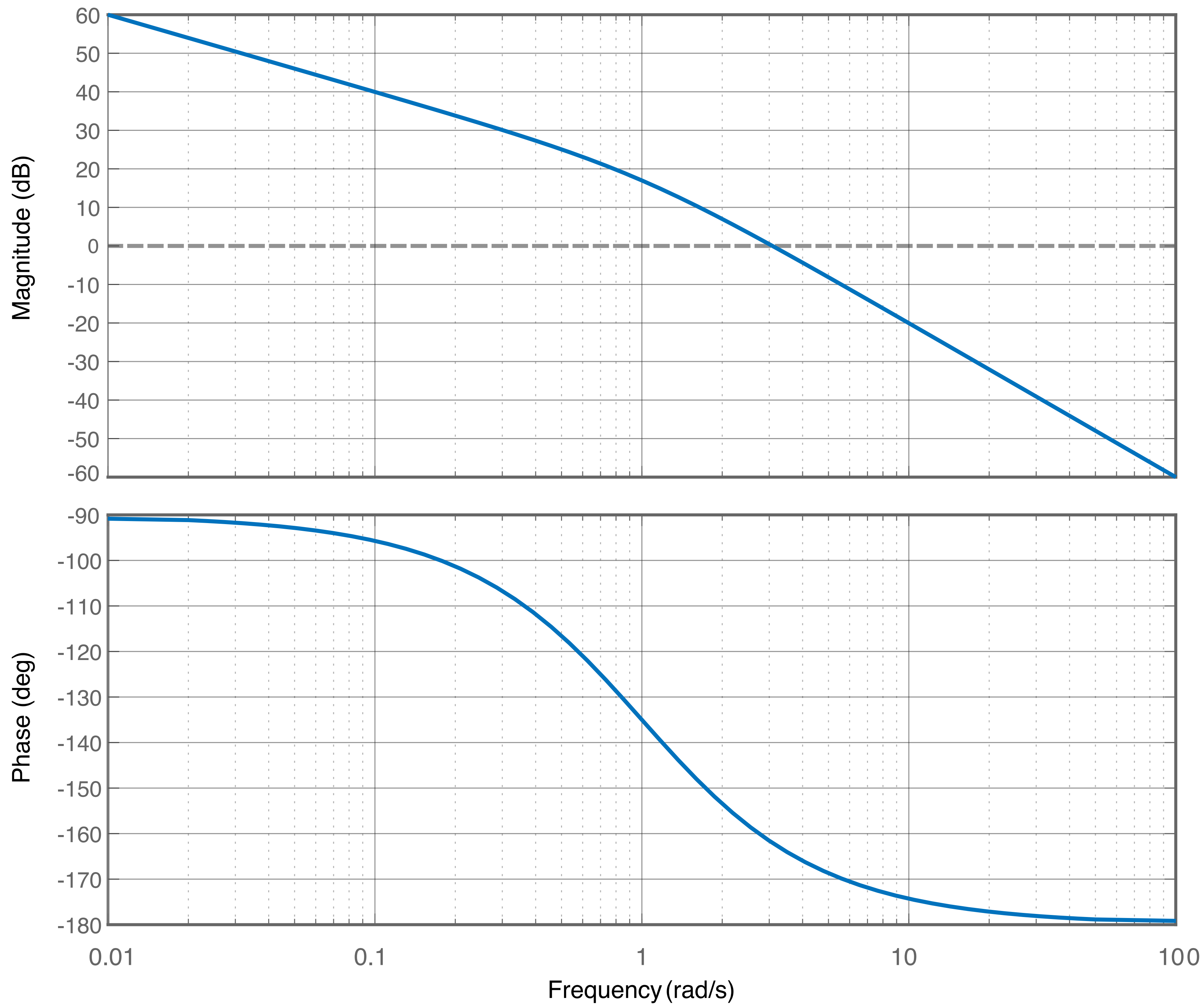
$$\bar{G}(s) = \frac{10}{s(s+1)}$$

$$G_c(s) = \frac{Ts + 1}{10Ts + 1}$$

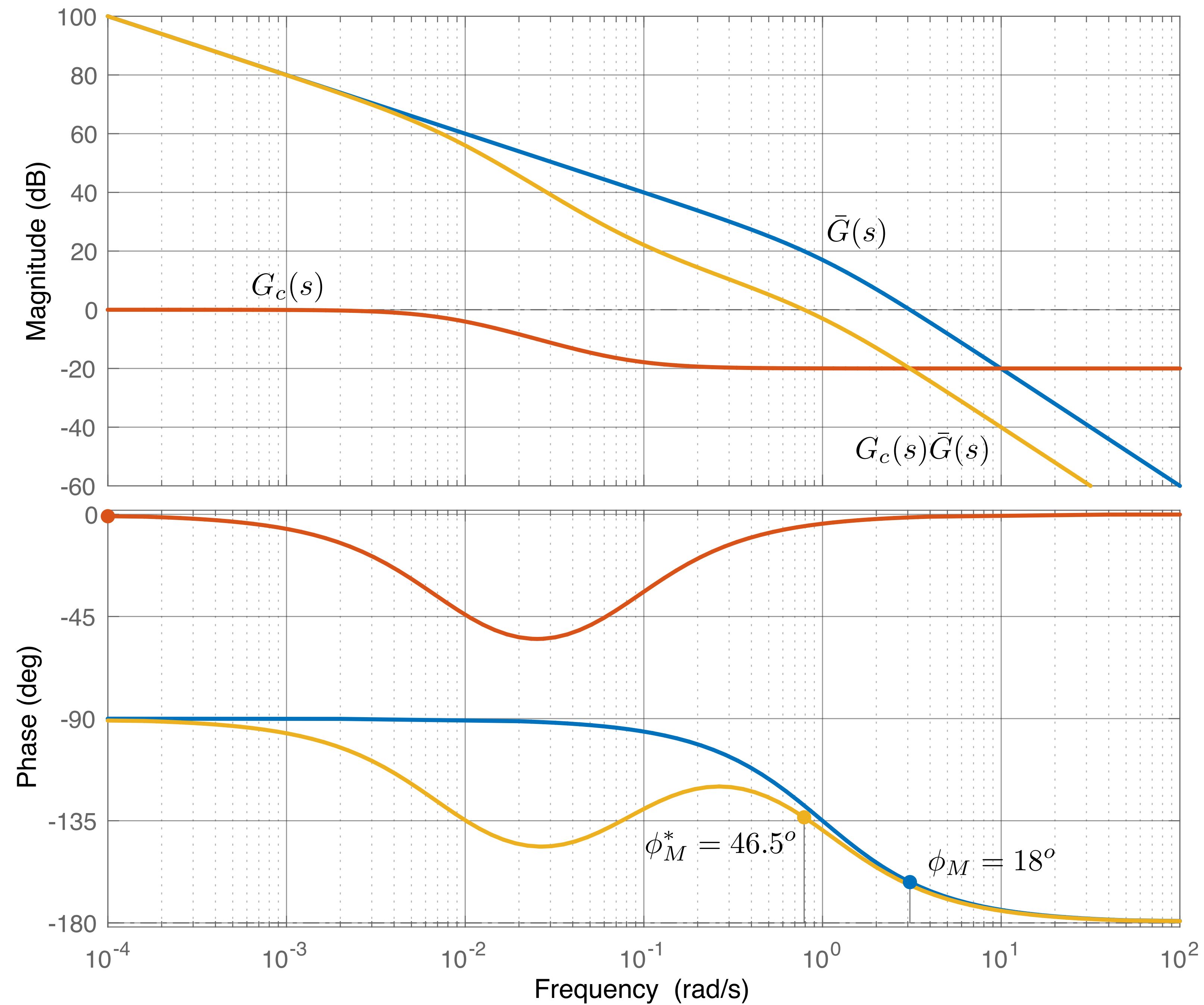


$$\bar{G}(s) = \frac{10}{s(s+1)}$$

$$G_c(s) = \frac{12.5s + 1}{125Ts + 1}$$



Results (Bode)



Results (Step Response)

